

ISMC121 ANALYSIS OF SYSTEM AND DESIGN

Final Exam

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Department of Computer Science

**RED CORNER MANAGEMENT INFORMATION SYSTEM
(RCMIS)**

Group 6

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RED CORNER MANAGEMENT INFORMATION SYSTEM

1 PROJECT DESCRIPTION

Red Corner Management Information System (RCMIS) is a management information system (MIS), executive support system (EES), and office automation system (OAS) that hopes to improve the operational efficiency of Red-Corner Hotel and Residences. This system will include the following features: booking management, semi-automated tenant utility billing, inventory management, employee scheduling, record management, and an operations dashboard populated with relevant data.

2 BACKGROUND OF THE PROJECT

Red-Corner Hotel and Residences is a 3-story building with 18 rooms, 4 commercial spaces, and a front desk lobby. The establishment's operations can be divided into two main categories: hotel and tenant.

Hotel operations include managing bookings and services for daily rate accommodations. This part of the business operation requires significant manpower and supplies, so many daily tasks are associated with this portion (cleaning, reception, etc.). The operations staff (mainly the receptionist) manages daily bookings from walk-in clients and online clients; the receptionist is responsible for communicating with the guests and relaying to the housekeeping needed additional tasks.

The current system makes use of software to manage the business. These include online platforms such as Agoda, Airbnb, and Facebook. Agoda and Airbnb are both online booking platforms. On the other hand, Facebook is a form of social media widely used in the Philippines. All in all, this softwares helps in making Red-Corner accessible and readily available to the public.

Tenant operations include managing the portion of the 18 rooms that are rented and the 4 commercial spaces rented by local businesses. Managing this portion of the operations is less demanding and requires a few monthly tasks such as maintenance, utility billing, and other similar tasks. Even though these tasks are performed monthly, they are done on different days of the month due to outside constraints such as the utility billing schedule, different tenant start dates (therefore different billing periods), and other similar constraints. These end as additional overhead tasks spread across the month, which can be difficult to manage due to the inconsistent nature of the tasks.

Data gathering revealed some issues experienced by the staff and the business owner. These problems include "double booking" (too many rooms booked than the business can accommodate caused by unsynchronized records), employee shift scheduling (sudden absentees), hostile guests (usually caused by double booking), dishonest staff, non-functioning guards, and maintenance of equipment.

Currently, business processes in Red-Corner are feasible and manageable but can be revised to improve customer and employee experience. Instances the issues mentioned above sometimes happen and negatively affect service quality.

3 SIGNIFICANCE OF THE PROJECT

This project aims to enhance and improve the current system of Red-Corner Hotel and Residences by creating a management information system (MIS). With this, the team of system analysts has identified possible benefits of the proposed system to the organization.

Automation is the first benefit of the proposed system as it helps in the efficiency and accuracy of business processes. It offers convenience to both the members of the staff and admin as well as the customers. With automation, the staff also experiences ease because of the operational efficiency and the admin ease of management and monitoring. This also leads to customer satisfaction as they are offered better service.

The second benefit of this system is in the notification and organization of data. Staff are alerted about important events such as upcoming reservations, room checkouts, and low inventory levels. Guests are also notified of updates and changes to their bookings. The system also creates a centralized platform for accessing relevant information including reservations, guest details, housekeeping status, and maintenance logs. These help in keeping records organized.

Accuracy is the last benefit of this system as synchronized records in one platform significantly reduce the possibility of double bookings. It minimizes errors assuring accurate guest information, up-to-date inventory of the different types of rooms, and real-time updates to reflect booked rooms. The accuracy of information and mitigation of double bookings are pertinent in preventing agitating guests from becoming hostile and improve operational efficiency by showing accurate inventory logs and ascertaining precise financial data.

Due to the above benefits, the system renders better service to customers. It makes sure that customers feel satisfaction before, throughout, and after their stay. Overall, the proposed system promotes increased operational efficiency, empowered staff, accurate financial data, and enhanced customer experience.

4 CURRENT SYSTEM

The current system uses different software such as Agoda, Airbnb, and Facebook to facilitate communication and booking. Aside from that, the business accepts walk-in clients. They are all required to fill out a registration form before checking into their rooms.

4.1 Business Processes

As mentioned above, the two main components of the business process for the establishment are divided into hotel and tenant.

Business processes for hotels include all activities that revolve around accommodating daily rate guests. This consists of the booking process (reservation, payment, and confirmation), preparing rooms (cleaning, and quality control), supply management (essential toiletries, drinking water, etc.), and other needed activities.

Business processes for tenant operations include billing tenants for rent and utility bills, overseeing maintenance, accomplishing covered repairs, managing leases, and other concerns.

4.2 Process Flow Diagrams of the current business processes

The following diagrams are current process flow diagrams of business operations (PFD)

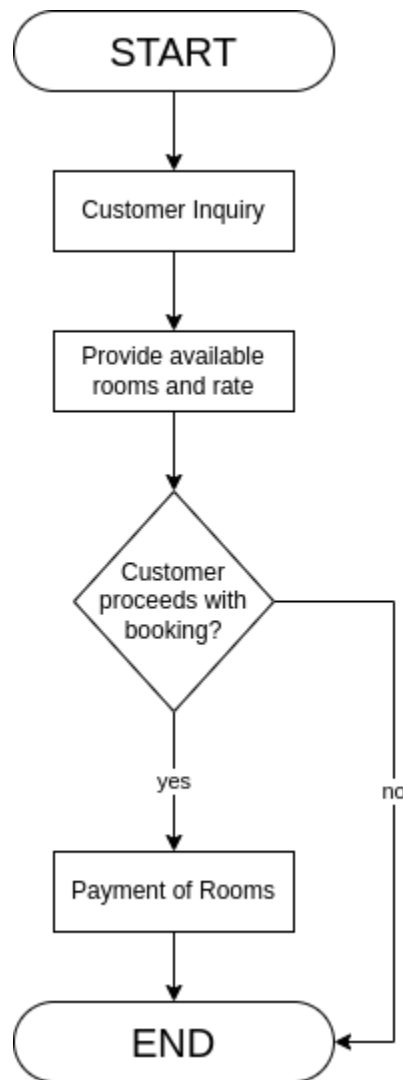


Figure 4.2.1 Current Booking PFD

A customer inquires about available rooms face-to-face, or through text, call, or Facebook page. The receptionist confirms how many rooms they need by asking about the number of guests, dates of stay, and other information such as room preference. The receptionist then proposes rooms and the corresponding price, for the client to decide to proceed with the booking. After payment, the booking is confirmed and the client may check in on their booked dates.

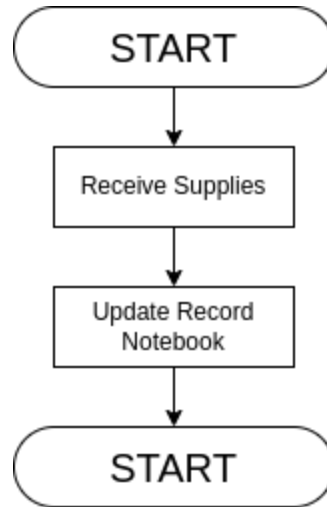


Figure 4.2.2 Current Inventory PFD

A supplier turns over supplies to the receptionist and updates the notebook for received items. The receptionist takes note of the quantity of supplies received and the date and signs the record on the notebook for the admin to check and review. Monitoring of supply is manual. When supply quantity looks low (i.e. not enough supply for the next month), the housekeeping staff notifies the receptionist, the receptionist notifies the admin staff, and finally the admin staff orders supplies.

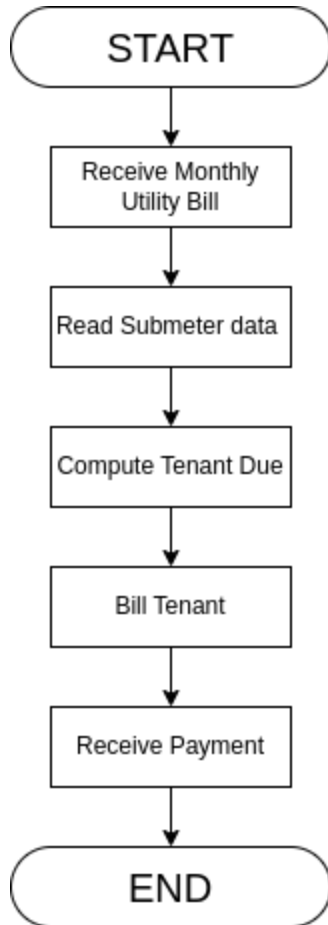


Figure 4.2.3 Current Tenant Billing PFD

Submeter data is recorded in a notebook based on the billing period found on utility bills. For example, given a billing period of December 12, 2024 to January 11, 2025, staff will take a picture of submeters between December 12, 2024 and January 11, 2025. When the bill arrives, the due amounts for all submeters are computed using a spreadsheet template, and tenants will be billed accordingly. The receptionist gives out the bill generated using the spreadsheet, complete with a due date and penalty for late payment. The receptionist will receive payment from tenants when tenants hand out payments at the front desk.

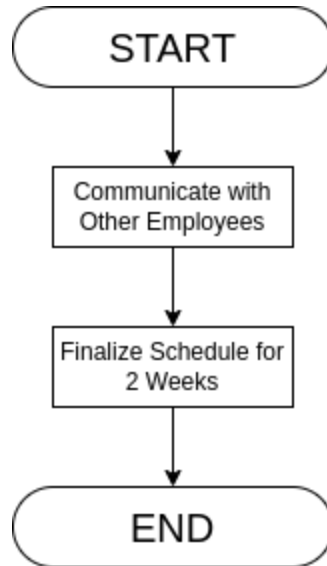


Figure 4.2.4 Current Booking PFD

The head staff contacts other employees and talks with them about their available dates for the next two weeks. After the schedule is finalized, the head staff sends out the finalized schedule to all staff and the admin for record purposes. The schedule will be implemented for the indicated dates.

5 PROPOSED SYSTEM

The following diagrams are proposed process flow diagrams (PFD) for RCMIS.

5.1 Process Flow Diagram

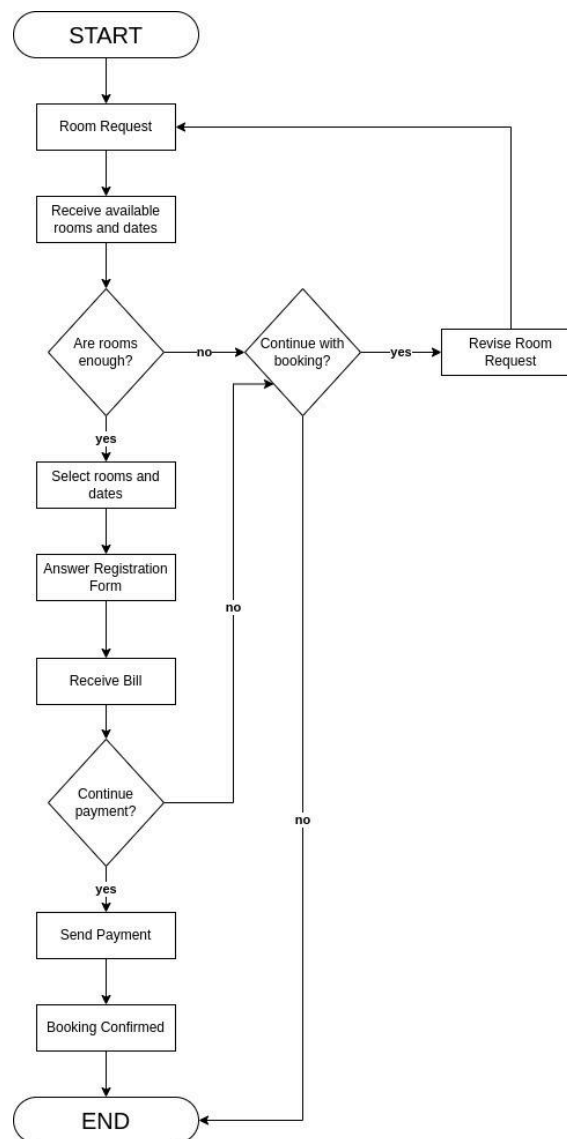


Figure 5.1.1 Proposed Booking PFD

The booking diagram illustrates the step-by-step procedure of the processes involved in making guest reservations inside the hotel. As the customer requests available rooms for the given dates, the client decides if the room availability fits to their decided booking option. If it is indeed available, they can select their desired room for the given date and proceed with payment options and after the payment has been sent, the booking they made will be confirmed and appointed. If the room's availability did not fit their request or if they did not proceed with the

payment, no reservations will be made and they will then be asked if they want to revise their room request or end the transaction instead.

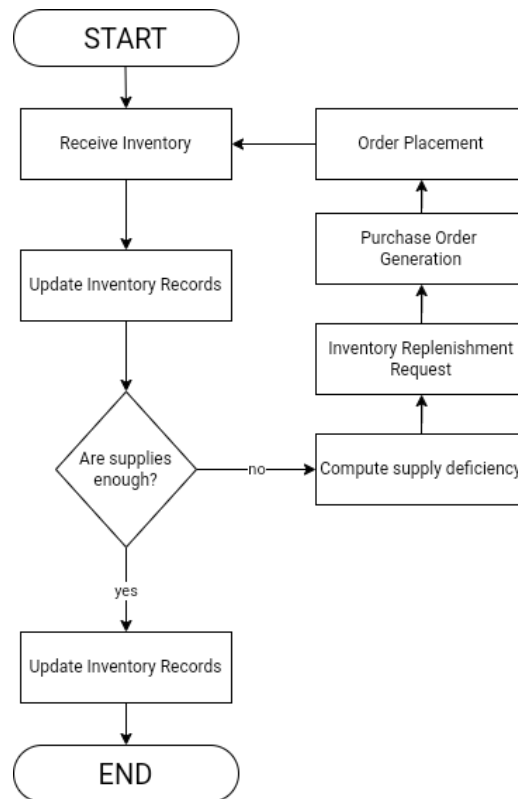


Figure 5.1.2 Proposed Inventory PFD

To keep track of the inventory inside the establishment, after receiving the supplies for the hotel, the inventory records are updated and decisions are made if the supplies bought are sufficient to conduct operations, if not, a new inventory replenishment request is made based on the computed supply deficiency and a purchase order will be generated to be included in the order placement.

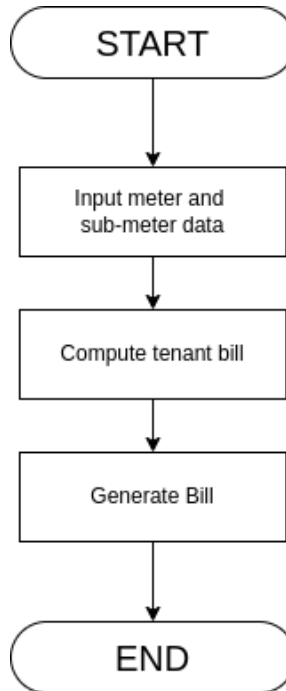


Figure 5.1.3 Proposed Tenant Billing PFD

For tenants who are staying for longer periods, their electricity and water bills are computed with the use of RCMIS. Utility meter and submeter data, and utility bill data are inputted in RCMIS, then it generates bills for all rooms and commercial spaces. The amount they consumed as seen in submeter data is factored in the computation for the billing statement generated.

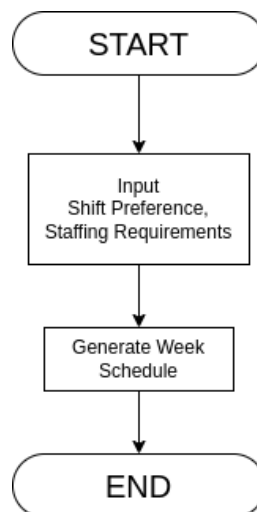


Figure 5.1.4 Proposed Scheduling PFD

The employee shifts are decided based on the employee's shift preference, availability, and staff requirements for the day. After the shift requirements are processed, a schedule for the week is generated.

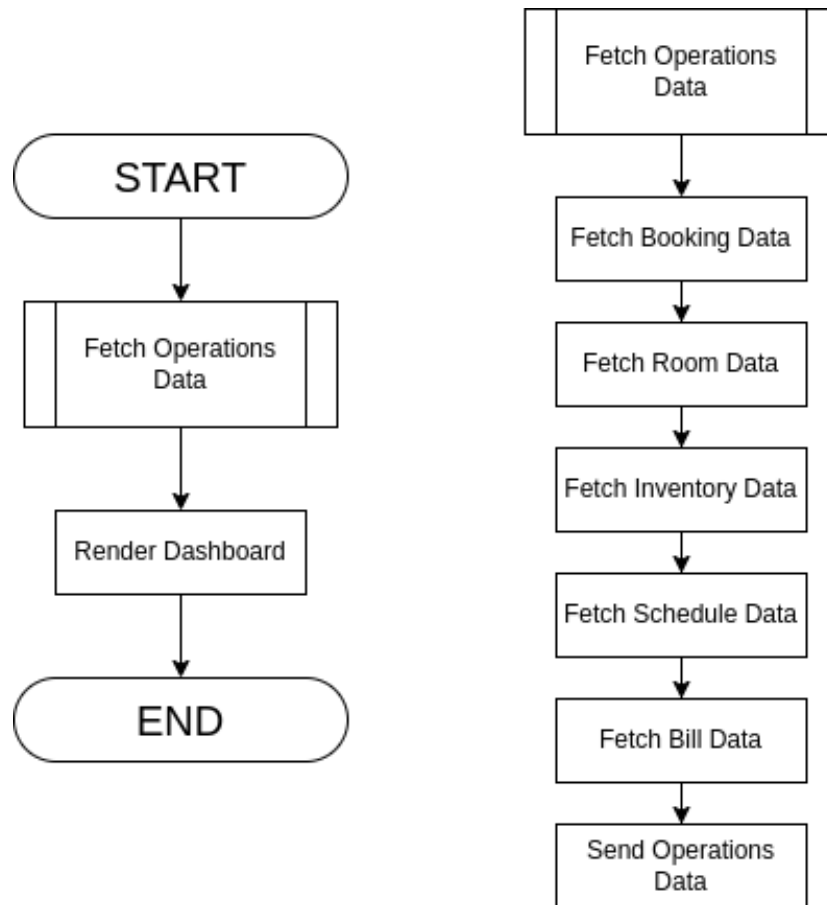


Figure 5.1.5 Proposed Dashboard PFD

To render the dashboard for analyzing the data included in hotel operations, the booking, room availability, inventory, employee scheduling, and bill data are compiled to generate the overall operations data.

5.2 Data Flow Diagram

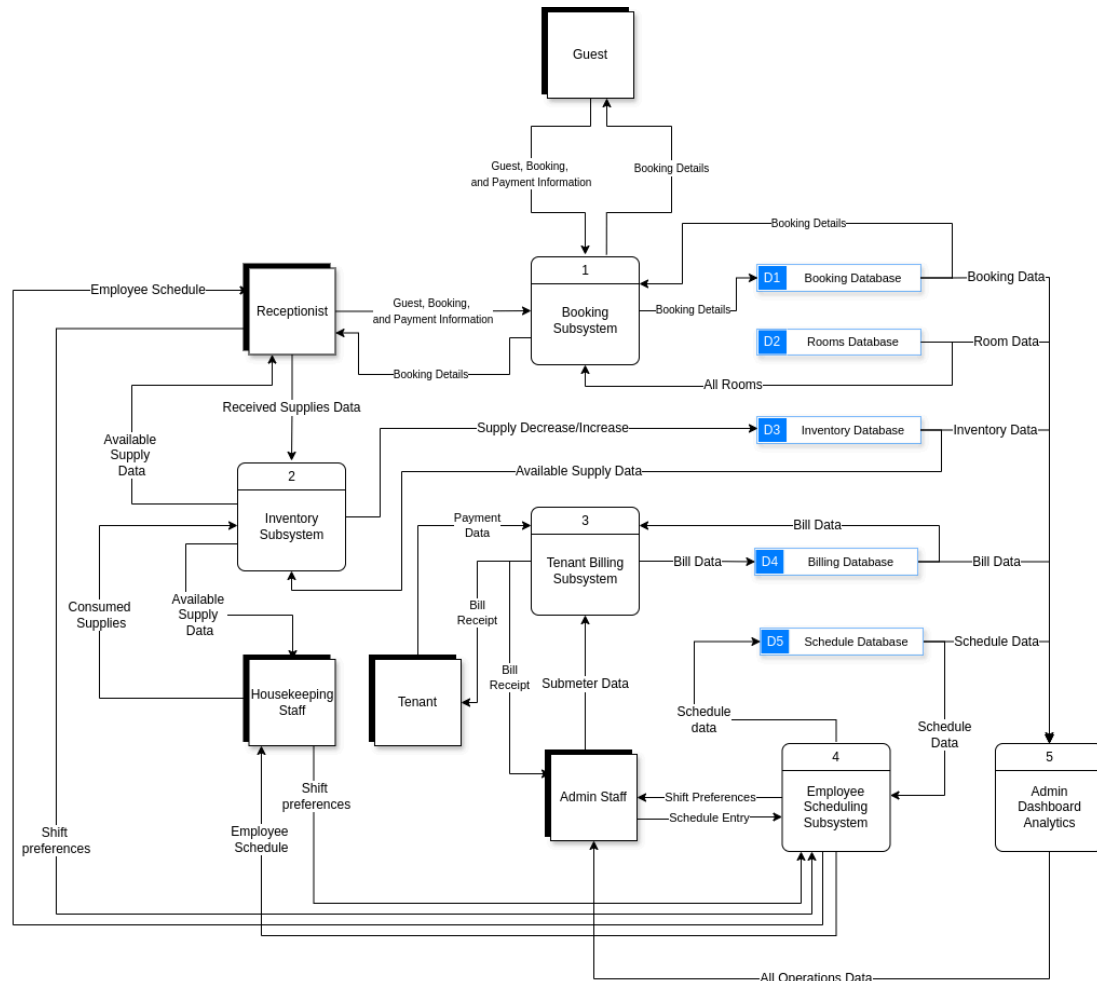


Figure 5.2.1 Proposed DFD for RCMIS

The data flow diagram illustrates the passing of information among entities involved in specific subsystems. It is seen in the diagram that there are 4 subsystems which are booking, inventory, tenant billing, employee scheduling, and admin dashboard analytics. These subsystems, excluding the admin dashboard analytics, each have their database to safely store data. The booking subsystem, however, has 2 databases that separate data for booking and for available rooms. In addition, the admin dashboard analytics is the fifth process that takes the data from all these databases. In the booking subsystem, the guest and the receptionist are the two entities that interact and receive information from one another. In the inventory subsystem, the receptionist cooperates with the supplier for the needed supplies to be used by the housekeeping staff. The tenant billing subsystem computes the total bill to be paid by the tenant according to the submeter data that is sent by the admin staff to the subsystem. In the employee scheduling subsystem, all staff can choose their preferred shifts which will be considered

when generating their weekly schedule. Then all of the data stored in each database passes through the admin dashboard analytics to render a dashboard for the admin.

5.3 CRUD Matrix

Activity	Customer	Receptionist	Supplier	Housekeeping	Tenant	Admin Staff
Booking	CRUD					RUD
Guest Registration	C	CR				RUD
Invoice	R	R	CRUD			CRUD
Inventory Record		RUD				R
Bill (tenant utilities and rent)	R				R	CU
Shift		CU		CU		RU
View Dashboard						R

Table 5.3.1 Proposed CRUD Matrix

The CRUD matrix above shows how different users interact with the specific tasks and activities involved in the system. For booking, the customer is able to create, read, update, and delete their booking while the admin staff has the same abilities with the exception of creating a booking. In the guest registration, the customer is able to create a registration as it is a required part of the process of booking. The receptionist is able to create the registration by encoding a record for the client as well as read the completed registration. The admin staff is allowed to read, update, and delete the guest registration as needed. For the invoice, which involves the payment, bill, and receipt, the customer and the receptionist are able to receive and read these data. On the other hand, the supplier and the admin staff are able to create, read, update, and delete as they see fit. For the inventory record, the receptionist is the main handler of this task so they are able to read, update, and delete data from this record while the admin staff is able to read the alterations made. In the bill, which includes tenant utilities and rent, both the customer and the tenant can read it as a form of invoice that they receive. The admin staff is the one that is responsible for creating and updating the bill. For the shifts of the staff, the receptionist and housekeeper are able to create and update the schedule they want while the admin staff is able to read and update it as needed. In the view dashboard, only the admin staff is able read the contents of it.

5.4 Functional and Non-Functional Requirements

Functional Requirements:

- Booking
 - Create new reservations with guest details, room selection, dates, and pricing.
 - Modify existing reservations.
 - Securely process various payment methods for reservations.
 - Confirm or cancel reservations.
- Inventory
 - Maintain a detailed inventory of all rooms in the hotel.
 - Track room types, amenities offered, and room status (available, occupied, dirty, under maintenance).
 - Update room status in real-time based on reservations, check-in/out, and housekeeping updates.
 - Track inventory of additional items offered (toiletries, etc).
 - Track item usage by room or guest.
 - Automatically update room inventory upon reservation confirmation and cancellation.
 - Reflect room status changes upon check-in and check-out processes.
- Tenant Billing
 - Access guest information and reservation details (room type, dates, rates).
 - Track additional charges incurred by the guest during their stay.
 - Generate accurate bills for guests upon check-out, including room charges, additional charges, taxes, and fees.
 - Seamless integration with the booking subsystem to automatically populate billing details based on reservations.
 - Generate reports on guest spending habits, revenue generated by different services, and outstanding balances.
 - Provide insights for data analysis and identifying potential revenue streams.
- Employee Scheduling
 - Maintain employee profiles with contact information, roles, skills, and availability preferences.
 - Track employee qualifications, certifications, and training records.
 - Create and manage employee schedules for different departments.
 - Assign tasks and shifts based on employee skills, availability, and department needs.
 - Allow for flexible scheduling options (full-time, part-time, etc.).
 - Accommodate employee requests for time off, vacations, or shift swaps.
 - Manage staff leave and track remaining balances.
 - Generate reports on employee work hours, overtime, and labor costs.
- Admin Dashboard

- Display key performance indicators (KPIs) relevant to the hotel business on a user-friendly dashboard.
- Include visualizations like charts, graphs, and tables for easy interpretation of data.
- Allow customization of the dashboard to display preferred metrics and timeframes (daily, weekly, monthly, yearly).
- Provide insights into various aspects of the hotel business.

Non- Functional Requirements:

- Booking
 - Secure storage of guest data following data privacy regulations.
 - Encryption of sensitive information like payment details.
 - Fast response times for search queries and reservation processing.
 - Ability to handle high volumes of booking requests during peak seasons.
 - User-friendly interface for both guests and hotel staff.
 - The booking system should be accessible 24/7 for guests to make reservations online.
 - Generate reports on booking trends, occupancy rates, and revenue generation.
- Inventory
 - Maintain real-time and accurate data on room availability and item stock levels.
 - Minimize discrepancies between system data and physical inventory through regular audits.
 - Seamless integration with the booking subsystem to ensure synchronized updates on room availability.
 - Secure access control to prevent unauthorized modification of inventory data.
 - Fast response times for room availability checks and inventory updates.
 - Ability to handle high volumes of booking requests and check-in/out processes.
 - User-friendly interface for staff to manage room and item inventory efficiently.
 - Generate reports that are clear and customizable.
- Tenant Billing
 - Ensure accurate calculations for room rates, taxes, and additional charges.
 - Minimize billing errors through automated processes and data validation.
 - Secure storage of guest financial information following data privacy regulations.
 - Secure user authentication and access control for staff authorized to view and manage billing data.
 - Encryption of sensitive information like credit card details during transactions.

- Fast processing of billing transactions and check-outs to avoid guest wait times.
- Ability to handle high volumes of billing requests during peak seasons.
- User-friendly interface for staff to efficiently manage guest bills, payments, and accounts.
- Ability to accommodate different billing methods and payment options.
- Maintain a clear audit trail for all billing transactions and account activity.
- Allow for easy tracking and reconciliation of payments and charges.
- Employee Scheduling
 - Secure access control to prevent unauthorized modification of employee schedules.
 - Maintain data privacy for employee information, including contact details and availability preferences.
 - Efficient scheduling creation and management to minimize time spent on manual processes.
 - Ability to handle changes and updates to schedules in a timely manner.
 - User-friendly interface for managers and supervisors to create, edit, and view employee schedules.
 - Allow for customization of schedule templates for different departments and roles.
 - Accommodate various work schedules (fixed, rotating, on-call).
 - Provide clear and transparent visibility of employee schedules for both managers and staff.
 - Enable easy communication of schedule changes or updates to all relevant parties.
- Admin Dashboard
 - Restrict access to sensitive data based on user roles and permissions within the MIS.
 - Secure data storage and transmission following data privacy regulations.
 - Generate reports and visualizations efficiently to avoid delays in data analysis.
 - The dashboard should be responsive and load quickly for optimal user experience.
 - The system should be able to handle increasing data volume as the hotel business grows.
 - Accommodate the addition of new data sources and metrics in the future.

5.5 User Stories

- Booking
 - As a customer, I can view detailed room information (photos, amenities, descriptions), so that I can make informed booking decisions.
 - As a customer, I can make reservations online, so that I can book my stay conveniently.
 - As a customer, I can receive confirmation emails with reservation details and cancellation policy, so that I have all the necessary information.
 - As a receptionist, I can search for reservations by guest name, confirmation number, or date, so that I can quickly locate guest information.
 - As a receptionist, I can search for reservations by guest name, confirmation number, or date, so that I can quickly locate guest information.
 - As a receptionist, I can process guest check-out , so that I can ensure a smooth departure experience.
 - As a receptionist, I can update room status in real-time (available, occupied, dirty, under maintenance), so that I can maintain accurate inventory and allocate rooms smoothly.
 - As an admin staff, I can analyze booking trends (peak seasons, popular types), so that I can make data-driven decisions to improve guest experience and revenue.
 - As an admin staff, I can generate reports on booking activity, revenue, and occupancy rates, so that I can track performance and make informed operational decisions.
- Inventory
 - As a receptionist, I can view the real-time status of all rooms (available, occupied, dirty, under maintenance), so that I can prioritize tasks and ensure efficient cleaning or maintenance scheduling.
 - As a receptionist, I can see the availability of different room types for specific dates, so that I can accurately answer guest inquiries and efficiently assign rooms during check-in.
 - As an admin staff, I can track the inventory levels of all rooms and room types, so that I can identify potential overbooking situations and adjust room rates accordingly.
 - As an admin staff, I can analyze historical inventory data to identify trends in room occupancy and guest preferences, so that I can optimize room allocation strategies and potentially adjust room type availability in the long term.
 - As an admin staff, I can generate reports on inventory usage and stock levels, so that I can make informed decisions about purchasing and restocking to minimize waste and optimize inventory management.

- Tenant Billing
 - As a customer/tenant, I can view my current bill with detailed charges for room rate, additional services, taxes, and fees, so that I can understand my final charges before check-out.
 - As a customer/tenant, I can securely settle my bill at check-out using various payment methods, so that I can conveniently complete the payment process.
 - As an admin staff, I can monitor outstanding guest balances, so that I can identify and address potential payment issues promptly to maintain healthy cash flow.
 - As an admin staff, I can access reports on overall revenue generated from room rates, additional services, and other charges, so that I can evaluate the hotel's financial performance and make informed business decisions.
- Employee Scheduling
 - As a staff member, I can view my work schedule electronically, so that I have easy access to my upcoming shifts and work hours.
 - As a staff member, I can view the availability of other employees and potentially swap shifts directly within the system, subject to manager approval, so that employees can find replacements and manage their schedules more collaboratively.
 - As a staff member, I can receive notifications about schedule changes or updates through the system, so that I am always informed about any adjustments to my work schedule.
 - As an admin staff, I can track employee work hours, overtime, and potential areas for optimization, so that I can ensure labor costs are managed effectively and identify opportunities for scheduling improvements.
 - As an admin staff, I can manage staff leave requests (sick leave, personal leave) and track remaining balances, so that I can maintain accurate leave records and ensure adequate staff availability.
- Admin Dashboard
 - As an admin staff, I can customize the dashboard to display preferred metrics and timeframes (daily, weekly, monthly, yearly), so that I can focus on the information most critical for strategic decision making.
 - As an admin staff, I can view key performance indicators (KPIs) relevant to the hotel business on a user-friendly dashboard, so that I can quickly gain insights into overall performance and identify areas needing attention.
 - As an admin staff, I can analyze occupancy rates and revenue generated from room rates, so that I can optimize pricing strategies and maximize room revenue.

- As an admin staff, I can access and analyze raw data from various hotel operations (reservations, billing, housekeeping), so that I can identify patterns and opportunities for improvement across different departments.

5.6 Use Case Diagram

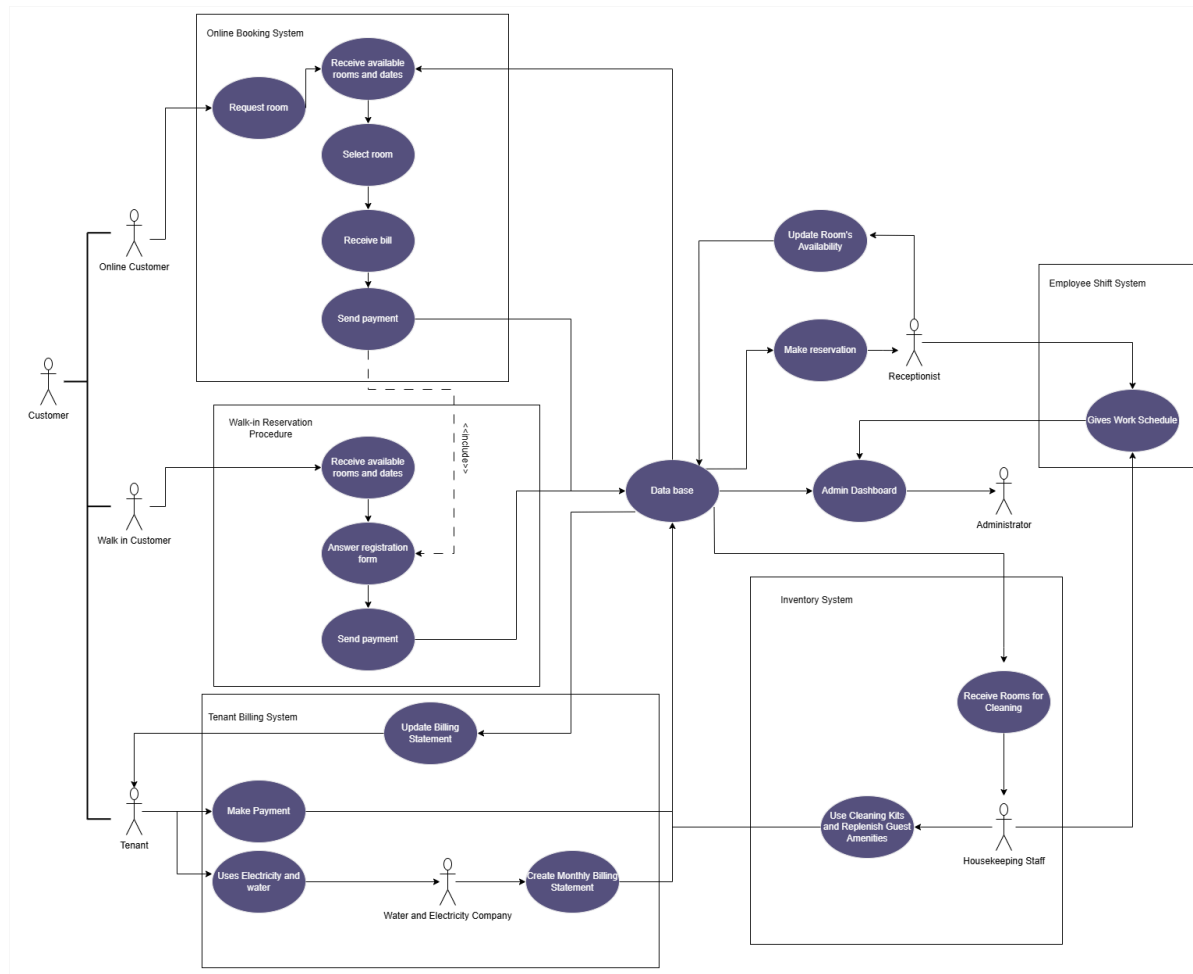


Figure 5.6.1 Proposed System Use Case Diagram

The only flaw that the current system has is that there are instances of double booking since the receptionist does not have direct access to the online booking system of the hotel. To address this issue the group decided to connect the walk-in and online bookings into a single centralized reservation database wherein the receptionist can easily access and mark the availed rooms and it lessens the workload of the general manager since he/she doesn't need to contact the receptionist every time an online booking is made. A tenant billing system, inventory system, and employee shift scheduling system are also integrated with the proposed system in order to ensure a smooth flow of information inside the hotel business.

5.7 Activity Diagram

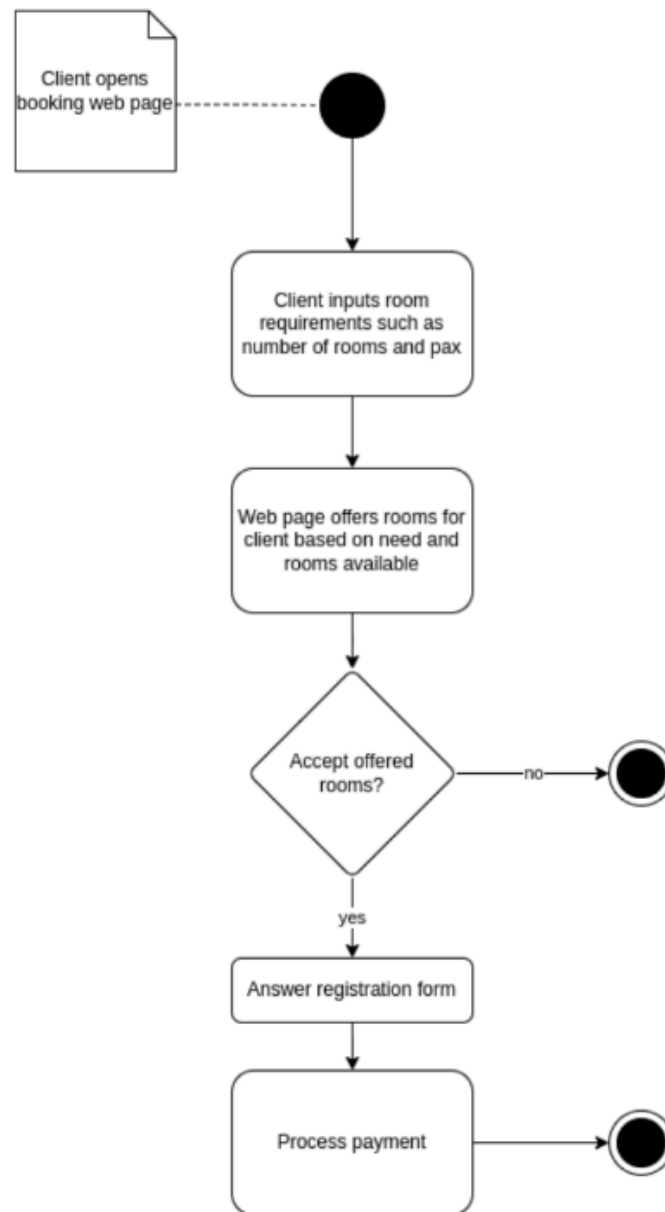


Figure 5.7.1 Proposed Booking Activity Diagram

This activity diagram showcases the process of the booking procedure for the customers. Although the diagram starts with the client opening the booking page, which implies that they are online customers, the rest of the procedure is applicable to walk-in customers as well. In the activity diagram, the customers are asked for their room requirements and are then offered rooms based on their needs. After that, if any of the suggested rooms satisfy their requirements, they are prompted to answer the registration form and process payment.

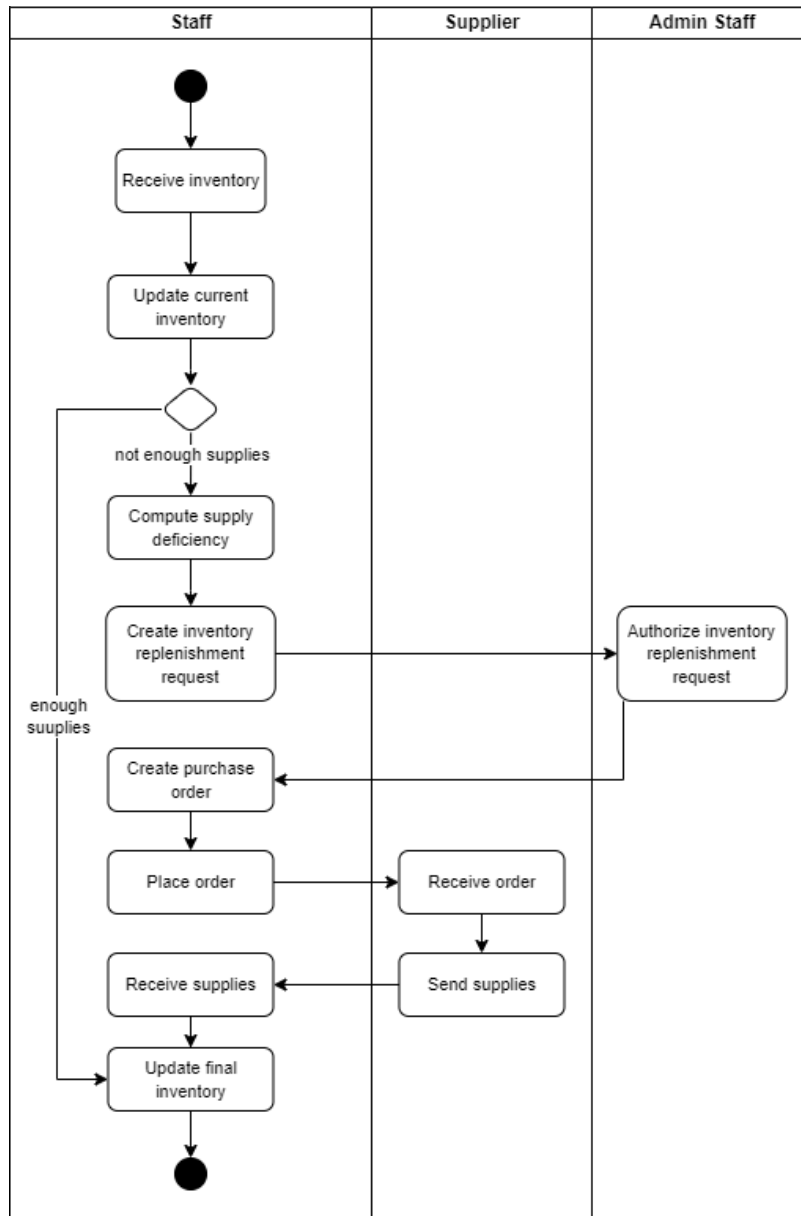


Figure 5.7.2 Proposed Inventory Activity Diagram

This activity diagram shows the whole process of updating the inventory. The staff, supplier, and the admin staff are the ones who are interrelated in this process. The staff updates the current inventory and checks if there are still enough supplies. If there is still enough inventory, the staff just updates it as the final inventory. If there are not enough supplies, the staff will check which supplies need replenishment and compute the total amount of supplies needed. Then a replenishment request will be sent to the admin staff to be authorized. Once it is authorized, the staff can then proceed to making and placing an order that will be sent to the supplier. The supplier will then send the supplies and when the staff receives the supplies, a final inventory update will be made.

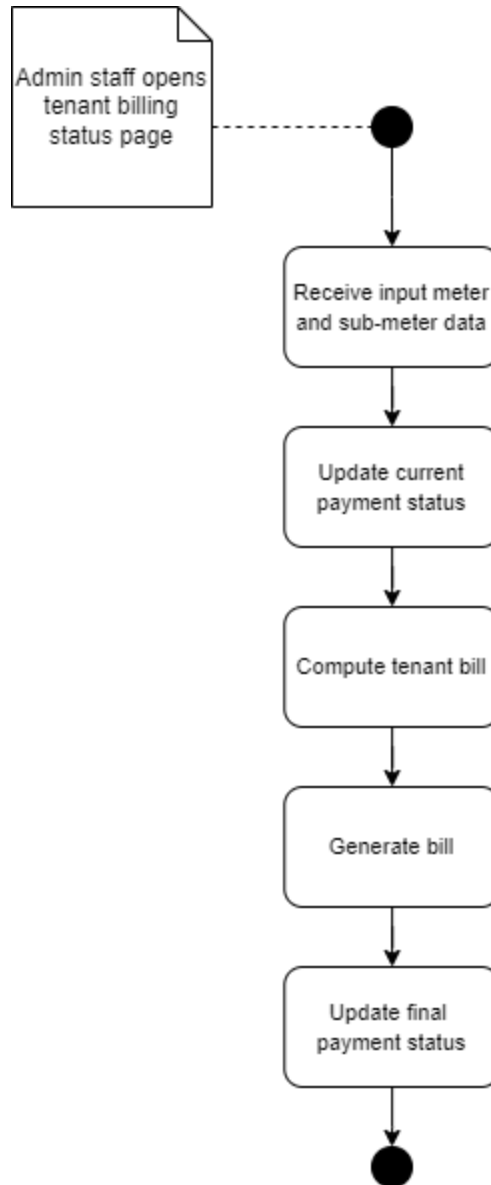


Figure 5.7.3 Proposed Tenant Billing Activity Diagram

The activity diagram above shows the updating of the tenant billing status. The admin staff opens the page and receives the input meter and sub-meter data. Then payment status is updated. Afterwards, the receive meter and sub-meter is used to compute the tenant bill which will be generated after. After the bill generation, the payment status will be updated again.

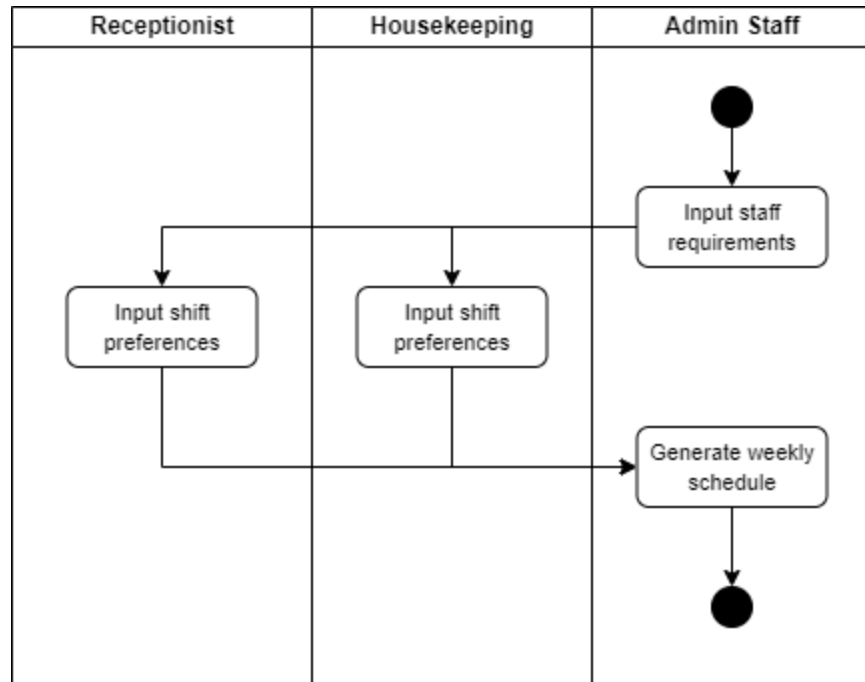


Figure 5.7.4 Proposed Employee Scheduling Activity Diagram

The activity diagram above shows how weekly schedules for employees are decided. The admin staff first input staff requirements for each day of the week. From these requirements, the receptionist and housekeeping can input their shift preferences. After that, the admin receives these preferences and compiles them into a weekly schedule.

5.8 Sample User Interface

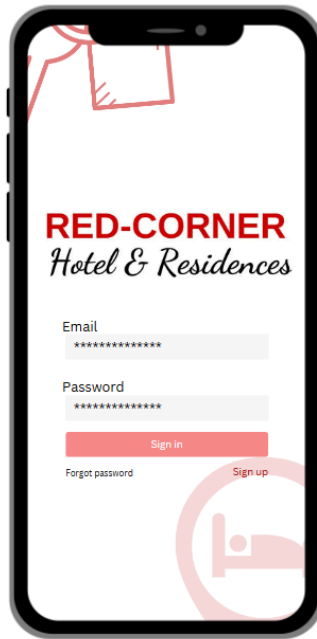


Figure 5.8.1 Log-in Screen

The log-in screen provides a security feature as well as an authentication feature, where users can access their accounts.

Buttons:

Sign in - The “sign in” button submits the inputted account information in order to access their accounts. If the login credentials are correct, it redirects the user to the home page.

Forgot Password - The “forgot password” button redirects the user to a verification page wherein they can reset their forgotten password given that they passed the security procedures.

Sign up - Provides a feature to let new users create their own account.

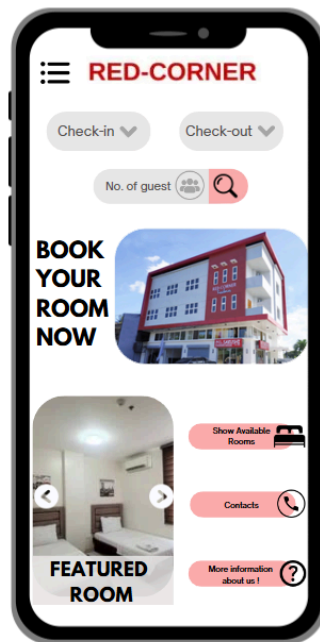


Figure 5.8.2 Home Screen

The home screen provides an overview of the features in the application wherein you can browse the featured rooms, or redirect to other screen pages.

Buttons:

Menu - This button redirects the user to their account settings.

“Check-in” - An expendable button that shows a calendar where users can select their start time and date for room reservations.

“Check-out” - An expendable button that shows a calendar where users can select their end time and date for room reservations.

“No. of guest” - A drop-down button that lets the user select the number of people that’s going to stay inside a room.

Search button - Submits the inputs from the “Check-in” , “Check-out“, and “No. of guest”. After verifying the availability of the rooms and the user requirements, it shows the user a list of rooms they can book for their own.

“Featured Room” - Redirects the user to the room’s details if they click on the image. A slider button is also added to let users glide through the hotel’s offers.

“Show Available Room” - Redirects the user to “Room Availability Screen”.

“Contacts” - Shows an information page where the contact details of the hotel are displayed.

“About us” - Shows an information page where the brief history, location, reviews, and features of the hotel are displayed.

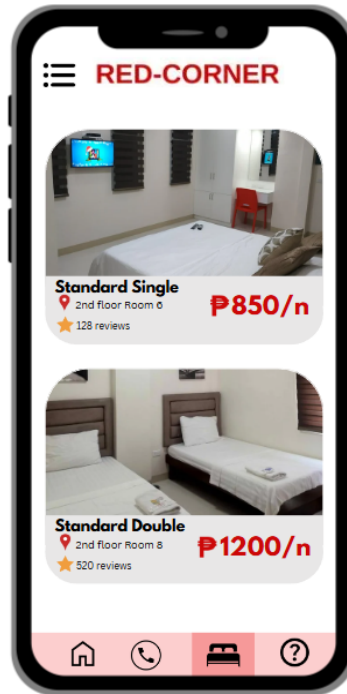


Figure 5.8.3 Room Availability Screen

The Room Availability Screen lets the user scroll through the available rooms and some of the basic details about them.

Buttons:

Home - Redirects the user to the home screen.

“Contacts” - Shows an information page where the contact details of the hotel are displayed.

About us” - Shows an information page where the brief history, location, reviews, and features of the hotel are displayed.

Room Displays - The images serve as a display and a clickable card that directs the user to the room’s details where they can scroll through the room’s information and make a reservation for it.

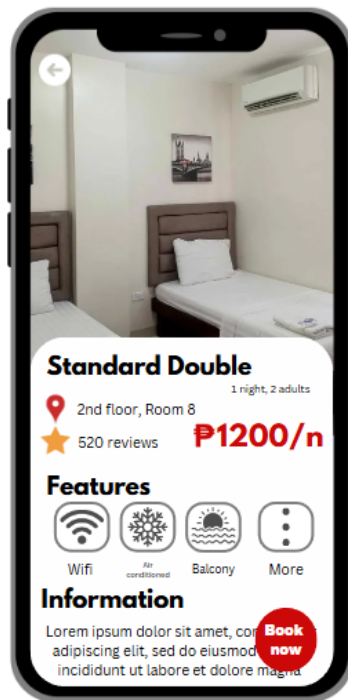


Figure 5.8.4 Room Information Screen

The Room Information Screen provides all the information about the user's selected room and grants them access to the booking button for that room.

Buttons:

Back - Redirects the user back to the Room Availability Screen.

Book Now - Directs the user to the Booking Details Screen.

RED-CORNER

←

Name:

Co-occupant:

Birth Date Contact Number

Valid ID Number

Stay Date

Su	Mo	Tu	We	Th	Fr	Sa
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

Form of payment
 ▼

Done

Figure 5.8.5 Booking Details Screen

The Room Information Screen provides all the information about the user's selected room and grants them access to the booking button for that room.

Buttons:

Stay Date - Let users select their time and date for room reservations.

"Form of payment" - A drop-down button that lets the user select their form of payment.

"Done" - Submits the inputs of the user and reserves the room selected at the given date.

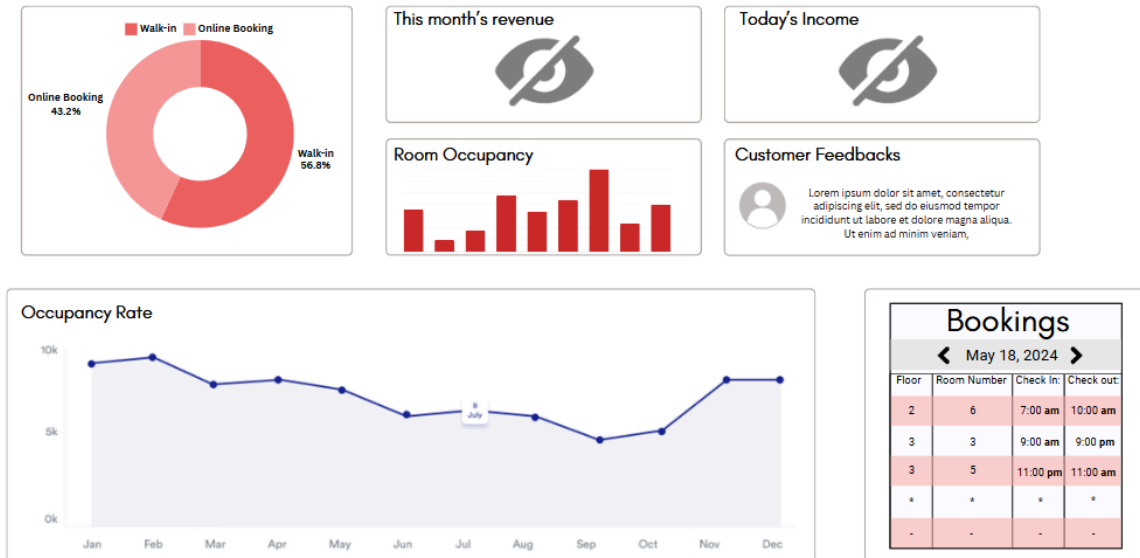


Figure 5.8.6 Admin Dashboard

The Admin Dashboard provides a real-time overview of key metrics and insights to help the admin efficiently manage hotel operations.

Elements:

Percentage of Walk-In and Online Bookings

- Displays a comparative analysis of bookings received through walk-ins versus online channels. It includes a pie chart or bar graph for a quick visual representation of the distribution.

This Month's Revenue

- Shows the total revenue generated for the current month. This figure is updated in real-time to reflect the financial status of the hotel.

Today's Income

- Allows for monitoring daily financial progress and making informed decisions on daily operations and promotions.

Occupancy Rate

- Measures the hotel's utilization rate, helping to identify trends and manage room inventory efficiently.

Room Occupancy

- Provides insights into customer preferences and popular room types, assisting in inventory management and targeted marketing.

Bookings

- A list of all current and upcoming bookings. This allows administrators to manage reservations effectively and ensure smooth operations.

Customer Feedback

- Collects and displays guest reviews and ratings.

Buttons:

Stay Date - Let users select their time and date for room reservations.

“Form of payment” - A drop-down button that lets the user select their form of payment.

“Done” - Submits the inputs of the user and reserves the room selected at the given date.

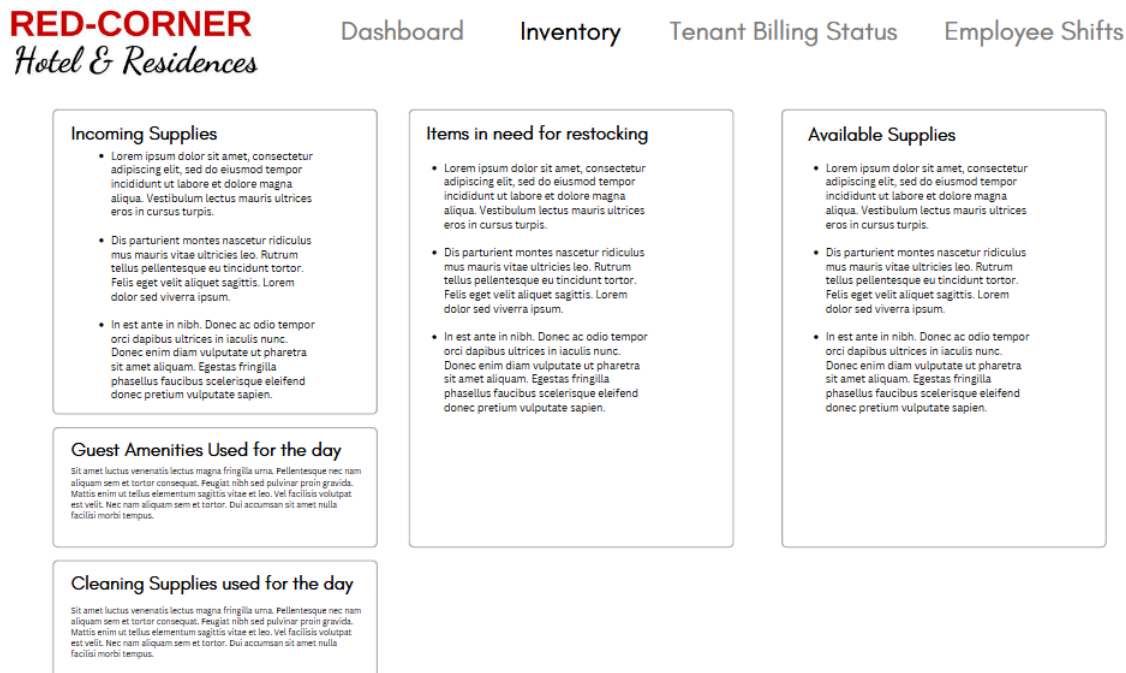


Figure 5.8.7 Admin Inventory Screen

The Admin Inventory Screen provides a detailed overview of inventory management, ensuring that all necessary supplies are tracked and maintained efficiently.

Elements:**Incoming Supply**

- Displays a list of supplies that are currently on order and expected to arrive. This includes details such as item names, quantities, estimated delivery dates, and supplier information.

Guest Amenities Used for the Day

- Shows the total amount of guest amenities used within the current day. This can include items such as toiletries, towels, bed linens, and complimentary items provided to guests.

Cleaning Supplies Used for the Day

- Details the quantity of cleaning supplies consumed during the day. This includes items like detergents, disinfectants, cleaning cloths, and other maintenance supplies.

Items in Need for Restocking

- Lists all inventory items that have fallen below the minimum threshold levels and need to be reordered. This section includes item names and recommended reorder quantities.

Available Supplies

- Provides an overview of the current inventory levels of all supplies available in the hotel.

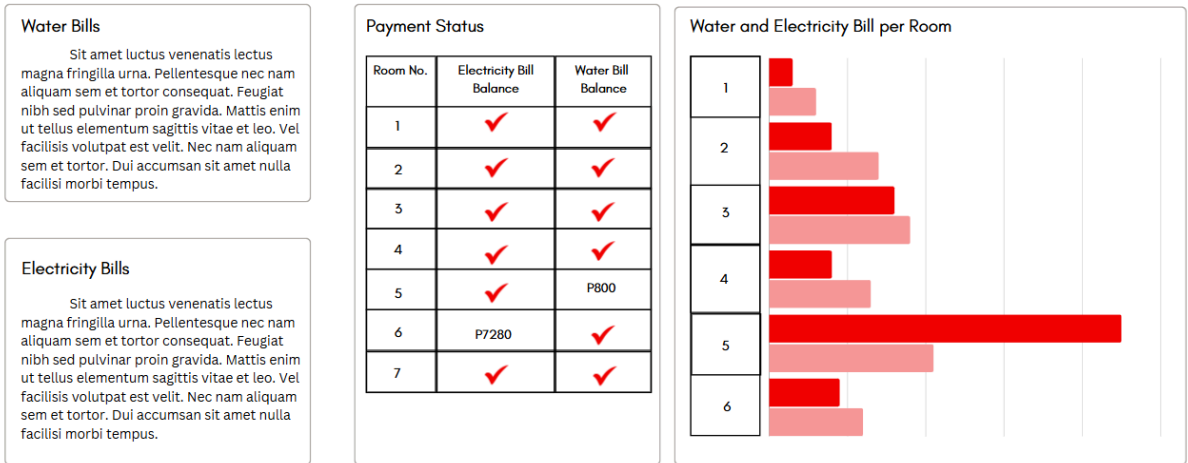


Figure 5.8.8 Tenant Billing Screen

The Tenant Billing Screen provides a detailed overview of billing and payment statuses for tenants.

Elements:

Water Bills

- Displays the current water bills for each tenant. This includes the amount due, billing period, and due date.

Electricity Bills

- Shows the electricity bills for each tenant. This section includes details such as the amount due, billing period, and due date

Payment Status

- Provides an overview of the payment statuses for all bills. This includes indicators for paid, pending, and overdue bills, along with payment dates and amounts.

Water and Electricity Bill per Room

- Offers a detailed view of utility usage on a per-room basis, helping in identifying any anomalies or unusually high consumption that may require attention.

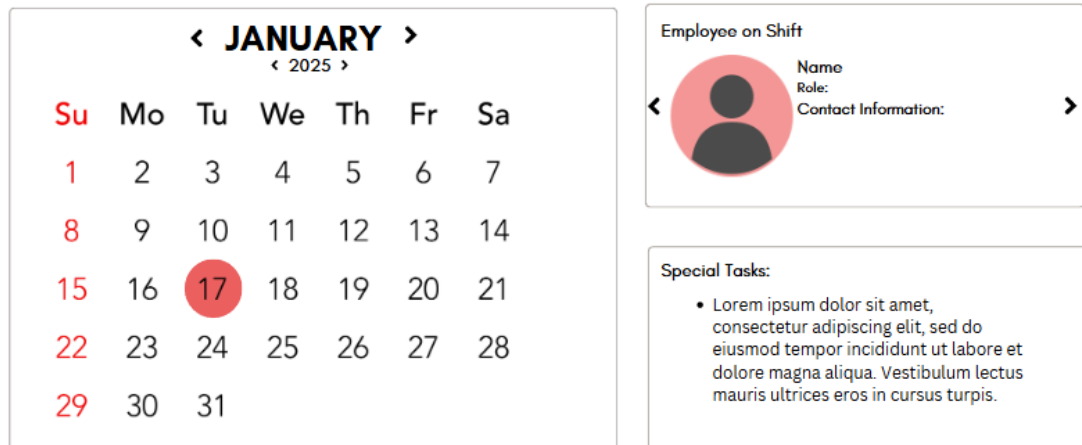


Figure 5.8.9 Employee Shifts Screen

The Employee Shift Screen provides an overview of the staffs' work schedule and their workload for a given day.

Elements:

Employee on Shift

- Enables administrators to quickly see who is working at any given time, ensuring that all areas of the hotel are adequately staffed and that shifts are running smoothly. This can also be used to address any immediate staffing needs or reassign employees as necessary.

6 APPENDIX

6.1 Interview Notes

Topic 1: General Operations (find out how the business operates)

What is the company's business model?

Who are your customers? Vacationers, boarders, and students reviewing for board exam

What are your plans for future improvements of the business? Fully computerized for control, less management time, efficiency. Improve interior design of the condo and branch out the business to Legazpi.

Topic 2: Availing products/services

How do clients avail of your services? Through their Facebook page, Agoda and Airbnb which are online booking platforms, and walk-in.

Challenges in terms of organizing:

- Schedule of client's stay - double booking
- Employee's time in and out - none

Other challenges:

- Hostile guests
- Dishonest staff
- Non-functioning guard
- Maintenance of equipments

Topic 3: Tasks performed by employees

How many employees do you have? 4 staff and 2 security

What are their tasks? Housekeeping, managing the front desk, and managing the facebook page and online booking platforms.

Other business compositions:

- Generator technician
- AC technician
- Laundry technician
- IT consultant for wifi
- Point person for cable subscription

Topic 4: Existing systems used

Do you use any software to manage your business? Yes. The online platforms like Agoda, Airbnb, and Facebook.

6.2 Guest Registration Form

RED-CORNER Hotel & Residences		REGISTRATION AGREEMENT		Registration No.: Nº 0499	
Room No.:	Room Rate:	No. of Person:	FDO:		
PLEASE PRINT IN BLOCK LETTERS					
Surname	First Name	M.I.		Mr.	Mrs. Ms.
Co-Occupant					
Profession		Company		Arrival Date / Time	
Permanent Address		E-mail Address		Departure Date / Time	
Nationality	Birthdate	Telephone Number		Form of Payment	
Valid ID Number	Place Issued	Date Issued		☐ Cash ☐ Bank Transfer ☐ G-Cash ☐ Agoda/Airbnb	
Visits: ☐ First Time ☐ Returning		REMARKS:			
BY SIGNING THE SPACE PROVIDED BELOW, I HEREBY AGREE TO THE FOLLOWING					
1. Payment is required in advance. 2. Check-in time is at 2 PM onwards on check-in date and check-out time is at 12NN on your check-out date. Early check-in and late check-out are subject to availability and with a charge of 100/hr. 3. Extra person will be charged accordingly. 4. Visitors staying beyond 10 PM will be charged as an extra person. 5. The management is not liable for any loss of your valuables not deposited with our front desk. 6. Personal belongings left or abandoned in the room shall be held for safekeeping for a period of thirty (30) days. After which the Management			7. shall be relieved of any responsibility in keeping the said belongings and shall be free to dispose or sell the same to cover unpaid accounts and expenses for safekeeping or storage. 8. Regardless of charge instructions, I- the guest- acknowledge that I am personally liable for the payment of all charges incurred by me during my stay at the RED-CORNER HOTEL and RESIDENCES. 9. Payment accepted is CASH (Philippine Peso), G-Cash, or Bank Transfer. 10. Senior Citizen/PWD ID must be presented upon check-in. 11. All suits arising out of this agreement shall be filed exclusively in Naga City and no other courts.		
Guest's Signature _____					
Dons St. corner Dayangdang St., Naga City, Camarines Sur 09285594538 / 09088960950					

6.3 Brochure

ROOM RATES per day

	Regular Rate	Promo Rate	
Standard Single	P1100.00	P850.00	1 Pax
Standard Double	P1500.00	P1200.00	2 Pax
Deluxe Queen	P1700.00	P1350.00	2 Pax
Deluxe Twin	P1700.00	P1350.00	2 Pax
Executive w/ Private Balcony	P1950.00	P1450.00	2 Pax
Extra Person w/o Bed	P200.00		
Extra Person w/ Bed	P300.00		

*Room rates are subject to change without prior notice.

* Actual Picture

AMENITIES

- Daily Housekeeping Service
- Secure Parking
- Hot & Cold Shower
- Fully - Airconditioned Rooms
- with FREE WIFI
- Cable TV
- Low Room Rates
- Accessible